

LaTeX Matrix Environments & Applications

Syntax, code tables, layout structuring, and practical uses across scientific research.

Setup & Matrix Syntax Rules

Required package

```
\usepackage{amsmath}
```

Required for the matrix environments used throughout this presentation.

LATEX CODE

```
\documentclass{article}
\usepackage{amsmath}    % matrix environments

\begin{document}
  % pmatrix, bmatrix, vmatrix, ...
\end{document}
```

& • column separator

Separates entries within a row.

\\ • row break

Ends one row and starts the next.

Math mode

Wrap expressions in $...$ or $\left[...\right]$.

Parentheses (pmatrix) & Brackets (bmatrix)

Parentheses matrix

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

LATEX CODE

```
\begin{pmatrix}  
1 & 2 \\  
3 & 4  
\end{pmatrix}
```

Square-bracket matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

LATEX CODE

```
\begin{bmatrix}  
a & b \\  
c & d  
\end{bmatrix}
```

3 × 3 Square Matrix (bmatrix)

VISUAL STRUCTURE

$$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$$

A square matrix has the same number of rows and columns.

LATEX CODE

```
\begin{bmatrix}
1 & 0 & 2 \\
3 & 4 & 5 \\
6 & 7 & 8 \\
\end{bmatrix}
```

The code directly describes the row/column structure seen in the rendered matrix.

Syntax pattern

Each row is separated by \\ and entries by &.

Determinants (vmatrix) & Norms (Vmatrix)

Determinant • vmatrix

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

LATEX CODE

```
\begin{vmatrix}  
a & b \\  
c & d  
\end{vmatrix}
```

Matrix norm • Vmatrix

$$\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

LATEX CODE

```
\begin{Vmatrix}  
a & b \\  
c & d  
\end{Vmatrix}
```

$n \times n$ Matrix (Extended from 3×3 Structure)

VISUAL STRUCTURE

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Ellipses and dots compactly represent a larger $n \times n$ structure.

LATEX CODE

```
\begin{bmatrix}
a_{11} & a_{12} & \cdots & a_{1n} \\
a_{21} & a_{22} & \cdots & a_{2n} \\
\vdots & \vdots & \ddots & \vdots \\
a_{n1} & a_{n2} & \cdots & a_{nn}
\end{bmatrix}
```

Why this matters

The same row/column syntax scales from small examples to general mathematical notation.

Matrix Types & Code Table

Matrix Type	Delimiters	LaTeX Environment	Primary Research Usage
Plain Matrix	None	<code>\begin{matrix}</code>	Raw tabular math alignment
Parentheses Matrix	() Round	<code>\begin{pmatrix}</code>	Vectors, coordinate transformations
Square Bracket	[] Square	<code>\begin{bmatrix}</code>	Linear algebra, engineering papers
Determinant	Single pipe	<code>\begin{vmatrix}</code>	Calculating matrix determinants
Matrix Norm	Double pipe	<code>\begin{Vmatrix}</code>	Vector space norms & bounds
Curly Braces	{ } Braces	<code>\begin{Bmatrix}</code>	Set theory & state space models

Choose the environment whose delimiters and semantics match the mathematical object being presented.

Piecewise Functions (cases Environment)

VISUAL RENDER

$$f(x) = \begin{cases} x^2, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Uses matrix syntax (& and \\) to align conditional mathematical statements.

LATEX CODE

```
f(x) = \begin{cases}
x^2, & \text{\text{if } } x \ge 0 \\
-x, & \text{\text{if } } x < 0 \\
\end{cases}
```

Key idea

The cases environment reuses the same column separator and row-break logic used by matrices.

Multi-Condition Functions (3-Case Signum Function)

VISUAL RENDER

$$\operatorname{sgn}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$

Multiple conditions are simply additional rows separated by `\\`.

LATEX CODE

```
\operatorname{sgn}(x) = \begin{cases}
1, & \text{\text{if } } x > 0 \\
0, & \text{\text{if } } x = 0 \\
-1, & \text{\text{if } } x < 0 \\
\end{cases}
```

Extension

The same alignment pattern scales naturally from two cases to three or more.

Augmented Matrix $[A \mid b]$ with One Continuous Vertical Line

VISUAL RENDER

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 4 & 11 \end{array} \right]$$

The `|` column specifier creates one continuous vertical separator.

LATEX CODE

```
\left[  
\begin{array}{cc|c}  
1 & 2 & 5 \\  
3 & 4 & 11  
\end{array}  
\right]
```

Why it is useful

Augmented matrices make systems of linear equations compact and visually explicit.

Systems of Simultaneous Equations (align*)

VISUAL RENDER

$$2x + 3y = 7$$

$$x - y = 1$$

The alignment marker & keeps the equals signs vertically aligned.

LATEX CODE

```
\begin{align*}  
2x + 3y &= 7 \\  
x - y &= 1 \\  
\end{align*}
```

Alignment logic

Place & immediately before the symbol you want aligned across lines.

Grouping Equations with Right Bracket & Equation Number

VISUAL RENDER

$$\left. \begin{aligned} 2x + y &= 10 \\ x - y &= 2 \end{aligned} \right\}$$

A right closing bracket groups the equations as one system.

LATEX CODE

```
\begin{equation}
\left.
\begin{aligned}
2x + y &= 10 \\
x - y &= 2
\end{aligned}
\right\}
\end{equation}
```

Equation numbering

The equation environment can attach a single equation number to the grouped structure.

Matrix Code Syntax for Document Structuring

1. Table Structuring (tabular)

Use column definitions and row separators to build structured tables.

LATEX CODE

```
\begin{tabular}{|c|c|}
Item & Price \\
\hline
Book & \$10
\end{tabular}
```

The | specifier creates continuous vertical column lines.

2. Math Grids (array)

Useful when mathematical values need a table-like grid.

LATEX CODE

```
\begin{array}{c|c}
x & y \\
\hline
1 & 2
\end{array}
```

Matrix syntax organizes mathematical tables and grid coordinates.

3. Multi-Line Math (split)

Break a long formula while keeping its alignment.

$$(a+b)^2 = a^2 + 2ab + b^2$$

LATEX CODE

```
\begin{split}
(a+b)^2 &= a^2 + 2ab \\
&+ b^2
\end{split}
```

Key Takeaways for Seminar Presentation

1

Use amsmath

Always include
`\usepackage{amsmath}`
in the LaTeX preamble.

2

Master `&` and `\\`

`&` separates columns
and `\\` separates
rows across environments.

3

Match delimiters

Choose `pmatrix`, `bmatrix`,
or `vmatrix`
based on the research
context.

4

Beyond matrices

Apply the same
syntax ideas
to `cases`, `tabular`, and `align`
structures.

Thank You!

Questions & Discussion



LaTeX Matrix Environments in Research Methodology & Scientific Writing